



The Reading Room

BOOK RECOMMENDER

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The Problem

Too Many Choices

Millions of books available — readers are overwhelmed and don't know where to start.

Poor Recommendations

Generic suggestions don't reflect individual taste, mood, or reading history.

Wasted Time

Readers spend more time searching than reading — and often give up on bad picks.

Our Solution

A personalised book recommender app that uses Machine Learning to suggest books based on genre, description similarity, and reading preferences.

2,004 books

1,004 GoodReads
1,000 Open Library

K-Means (k=6)

clustering algorithm

TF-IDF + PCA

NLP & dimensionality
reduction

Streamlit App

deployed recommender

The Data

2,004

Total Books
in the dataset

1,004

from GoodReads
web scraping

1,000

from Open Library
API

14

Features
per Book

After several iterations across different sources and scraping approaches, we settled on a combined dataset that gave us the richest and most complete data.

Fields collected per book:

Title · Author · Year Published · Genre · Description

Average Rating · Number of Ratings · Pages · ISBN

Cover Image URL · Language · Publisher · Series · Edition

Data Cleaning

Multi-language titles, authors & descriptions

Translated content from Arabic, Japanese, Greek, Persian and other scripts to English.

Low-resolution cover images

Detected small GoodReads thumbnails and replaced with larger, higher-quality versions.

Messy descriptions

Removed HTML tags, markdown links, line breaks, and irrelevant publisher metadata.

Duplicate & noisy author data

Cleaned repeated names and publisher info incorrectly listed in the author field.

Year & pages formatting

Extracted 4-digit years from free-text strings; standardised page counts to numeric.

Genre standardisation

Mapped 100s of genre variants to 66 clean categories using a custom mapping dictionary.

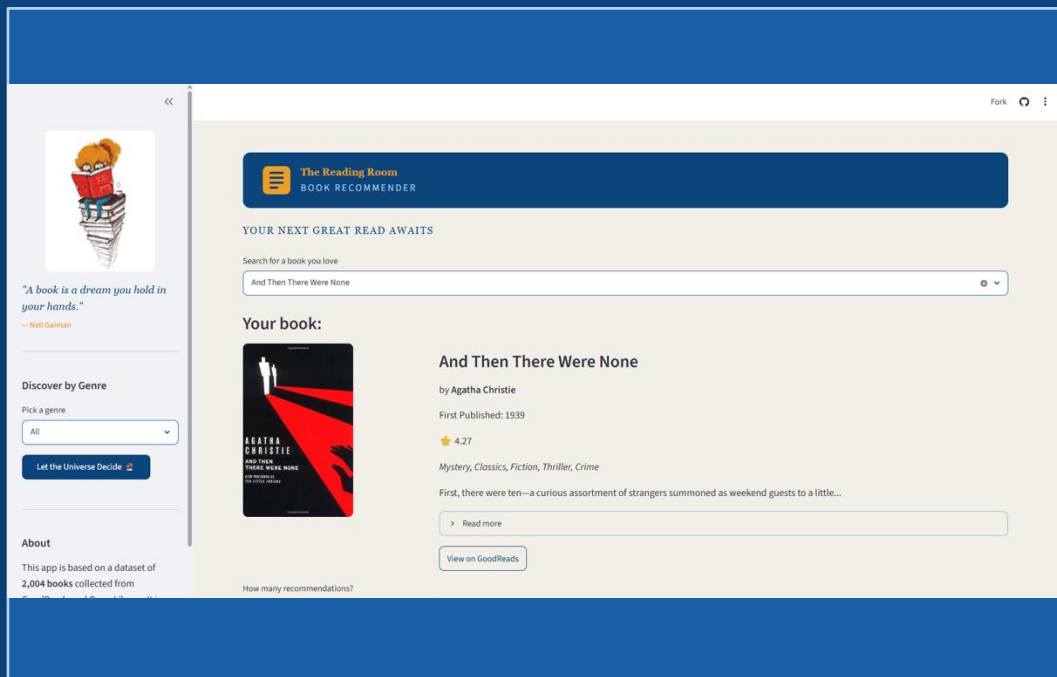
The ML Pipeline



Why k=6?

Silhouette Score: k=4 scored highest (0.179), k=6 scored second (0.166). We chose k=6 for richer, more diverse cluster themes — better for a book recommender experience.

The App



Key Features

Smart Book Search: search from 2,004 books across GoodReads and Open Library

Discover by Genre: random book picker filtered by genre

Adjustable Results: slider to choose from 1 to 9 recommendations

Book Details: cover image, rating, year published, genres and expandable description

Direct Links: one click to view any book on GoodReads or Google Books

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Results & Challenges

What Worked

- TF-IDF + PCA captured description similarity well
- Genre mapping standardised 100s of variants effectively
- K-Means k=6 produced diverse, meaningful clusters
- GoodReads + Open Library data complemented each other
- Streamlit app is intuitive and fast

Challenges

- Scoob blocked scraping — Next.js + Cloudflare 403
- Skeelo category pages rendered with JavaScript only
- Barnes & Noble, LibraryThing, Bookshop.org also blocked
- Many books had non-English titles & author names
- Low-resolution cover images needed systematic fixing
- Genre data was highly inconsistent across sources

Next Steps

01 Larger Dataset

Scale to 10,000+ books for more diverse recommendations across genres and languages.

02 Recommendation Feedback

Allow users to rate whether they would read a suggestion — use feedback to improve future recommendations.

03 Agglomerative Clustering

Test hierarchical clustering for more nuanced book groupings and compare against K-Means.

04 Richer NLP

Incorporate word embeddings (Word2Vec, BERT) for deeper semantic similarity beyond TF-IDF.



Thank You!

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